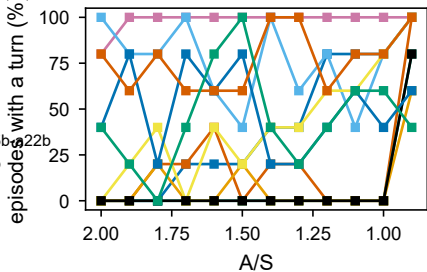


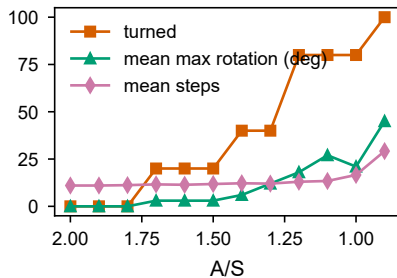
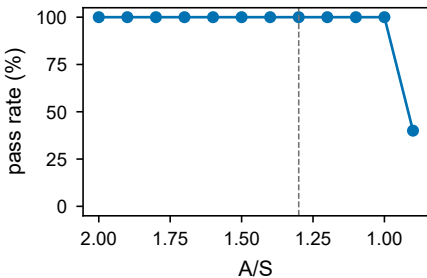
turn rate per aperture (flat = fixed policy)



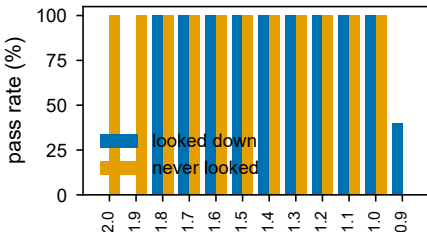
every model, sorted by overall pass rate (narrow = the two levels that need rotation: A/S 1.0 and 0.9)

	model	pass%	narrow	turn%	wide turn%	1st turn x	maxrot	steps	excess	look%	pass look	pass no look
claude-sonne		95	7/10	40	35	2.31	12	13.7	2.4	42	88	100
geminii-2.5-fl		93	6/10	7	0	7.06	4	12.6	1.3	8	60	96
deepseek-v4.1		92	5/10	18	11	3.77	4	13.3	2.0	53	84	100
glm-4.6v		92	6/10	8	4	5.00	6	13.1	1.7	10	50	96
gpt-4o		92	6/10	42	36	1.64	15	15.0	3.7	70	88	100
gpt-4o-mir		92	5/10	7	0	7.25	4	12.6	1.2	0	0	92
geminii-2.5-p		90	5/10	98	98	1.99	51	17.9	6.6	95	89	100
gpt-4.1		88	3/10	77	75	1.71	12	15.4	4.1	90	87	100
deepseek-v4		88	5/10	77	75	1.71	12	15.4	4.1	90	87	100
geminii-2.5-f		82	6/10	16	15	2.52	14	16.7	5.3	55	58	100
geminii-2.5-p		82	5/10	16	15	2.52	14	16.7	5.3	55	58	100

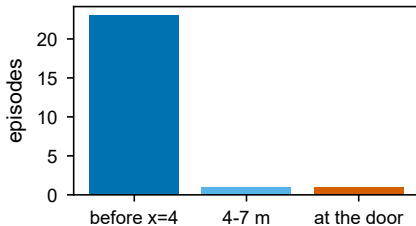
claude-sonnet-4-6 — 60 episodes, overall pass 95% outcome vs aperture



self-inspection (25/60 episodes) and outcome



when it looked down (mean x 1.9 m)



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.0	0	0.0	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.2	0	0.2	20	100	100
1.7	0.969	5/5	20	0	11.6	3	0.6	20	100	100
1.6	0.912	5/5	20	0	11.4	3	0.4	20	100	100
1.5	0.855	5/5	20	0	11.8	3	0.8	60	100	100
1.4	0.798	5/5	40	0	12.2	6	1.2	60	100	100
1.3	0.741	5/5	40	20	12.0	12	1.0	20	100	100
1.2	0.684	5/5	80	0	13.0	18	2.0	60	100	100
1.1	0.627	5/5	100	0	13.4	27	2.7	60	100	100
1.0	0.570	5/5	100	0	16.8	45	5.6	100	100	100

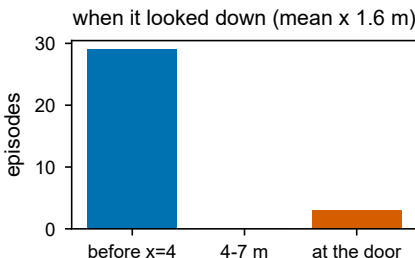
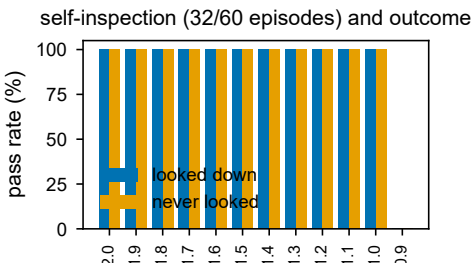
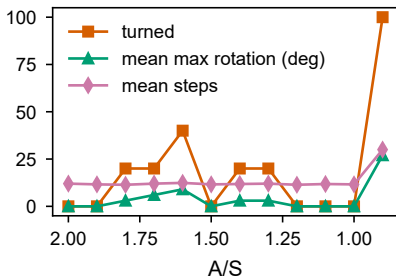
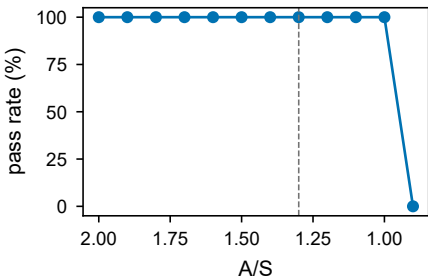
reasoning mentions (share of episodes): opening 100%, align 100%, turn 92%, width 70%, sideways 40%, blocked 28%, bow 13%, left 3%, look_down 4%, turn_right 3%, right 2%

strategy labels: FRONTAL 60%, ROT_EARLY 35%, ROT_MID 3%, ROT_LATE 2%

turned at all in 40% of episodes (first turn at x = 2.3 m on average); at apertures needing no turn it turned 39%

pass rate when it inspected itself 88% vs 100% when it never did

deepseek-v4.1-flash — 60 episodes, overall pass 92% outcome vs aperture behaviour vs aperture



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	12.0	0	1.0	80	100	100
1.9	1.083	5/5	0	0	11.6	0	0.6	60	100	100
1.8	1.026	5/5	20	0	11.4	3	0.4	20	100	100
1.7	0.969	5/5	20	0	12.0	6	1.0	60	100	100
1.6	0.912	5/5	40	0	12.4	9	1.4	60	100	100
1.5	0.855	5/5	0	0	11.6	0	0.6	60	100	100
1.4	0.798	5/5	20	0	11.8	3	0.8	40	100	100
1.3	0.741	5/5	20	0	12.0	3	1.0	20	100	100
1.2	0.684	5/5	0	0	11.4	0	0.4	40	100	100
1.1	0.627	5/5	0	0	11.8	0	0.6	60	100	100
1.0	0.570	5/5	0	0	11.6	0	0.6	40	100	100
0.9	0.513	0/5	0	100	30.0	27	15.0	100	0	-

reasoning mentions (share of episodes): opening 100%, align 100%, width 80%, turn 43%, blocked 32%, sideways 18%, depth 3%

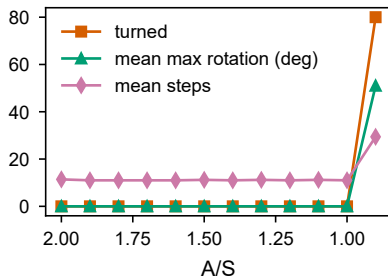
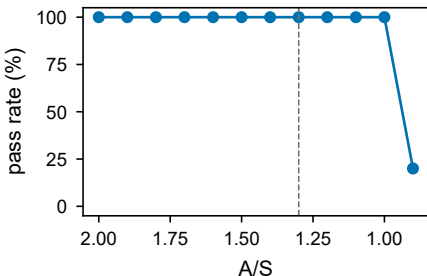
actions: forward 6%, look down 5%, turn_left 3%, turn_right 2%, right 2%

strategy labels: FRONTAL 82%, ROT_EARLY 10%, ROT_LATE 8%

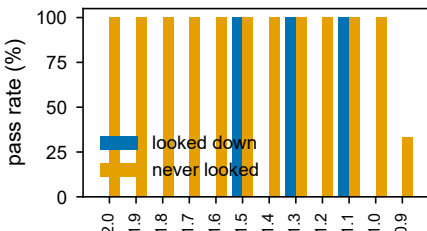
turned at all in 18% of episodes (first turn at x = 3.8 m on average); at apertures needing no turn it turned 100%

pass rate when it inspected itself 84% vs 100% when it never did

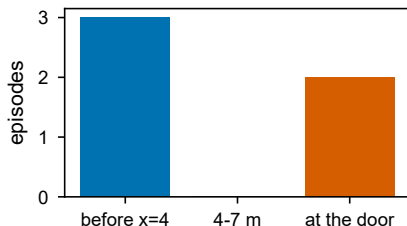
gemini-2.5-flash — 60 episodes, overall pass 93% outcome vs aperture



self-inspection (5/60 episodes) and outcome



when it looked down (mean x 4.0 m)



made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.4	0	0.4	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.0	0	0.0	0	—	100
1.7	0.969	5/5	0	0	11.0	0	0.0	0	—	100
1.6	0.912	5/5	0	0	11.0	0	0.0	0	—	100
1.5	0.855	5/5	0	0	11.2	0	0.2	20	100	100
1.4	0.798	5/5	0	0	11.0	0	0.0	0	—	100
1.3	0.741	5/5	0	0	11.2	0	0.2	20	100	100
1.2	0.684	5/5	0	0	11.0	0	0.0	0	—	100
1.1	0.627	5/5	0	0	11.0	0	0.0	0	—	100
1.0	0.570	5/5	0	0	11.0	0	0.0	0	—	100
0.9	0.513	0/1	0	0	11.0	0	14.4	40	0	33

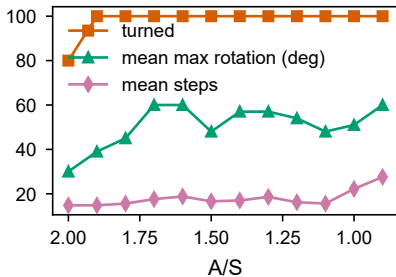
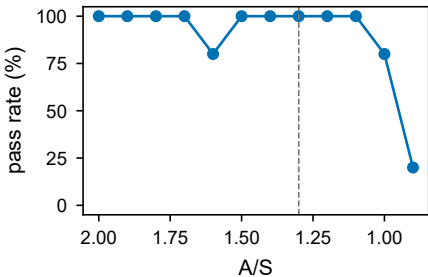
reasoning mentions (share of episodes): opening 100%, align 92%, turn 32%, blocked 22%, width 42%, sideways 38%, turn_right 3%, turn_left 2%, look_down 2%, left 1%

strategy labels: FRONTAL 93%, ROT_LATE 5%, ROT_MID 2%

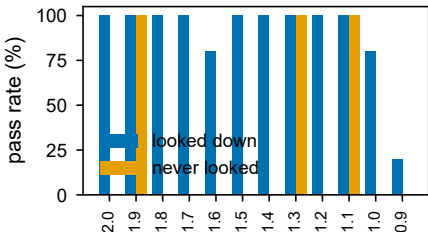
turned at all in 7% of episodes (first turn at x = 7.1 m on average); at apertures needing no turn it turned 0%

pass rate when it inspected itself 60% vs 96% when it never did

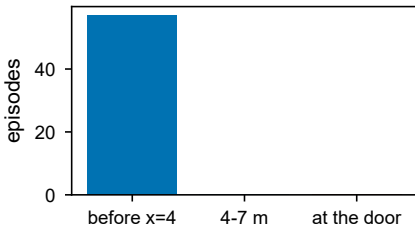
gemini-2.5-pro — 60 episodes, overall pass 90% outcome vs aperture behaviour vs aperture



self-inspection (57/60 episodes) and outcome



when it looked down (mean x 1.6 m)



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	80	40	14.8	30	3.8	100	100	—
1.9	1.083	5/5	100	40	14.8	39	3.8	80	100	100
1.8	1.026	5/5	100	60	15.6	45	4.6	100	100	—
1.7	0.969	5/5	100	80	17.6	60	6.6	100	100	—
1.6	0.912	4/5	100	75	18.8	60	7.8	100	80	—
1.5	0.855	5/5	100	60	16.6	48	5.6	100	100	—
1.4	0.798	5/5	100	60	17.0	57	6.0	100	100	—
1.3	0.741	5/5	100	40	18.6	57	7.6	80	100	100
1.2	0.684	5/5	100	80	16.2	54	5.2	100	100	—
1.1	0.627	4/5	100	30	22.2	51	11.2	100	80	—
1.0	0.570	0/5	100	100	27.6	80	12.6	100	20	—

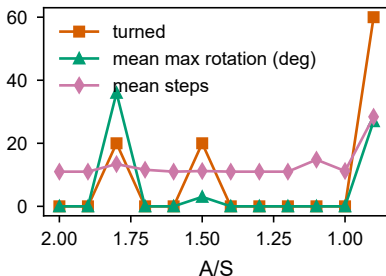
reasoning mentions (share of episodes): opening 100%, width 100%, turn 98%, sideways 98%, align 95%, blocked 27%, align 23%, right 21%, look_down 9%, turn_left 7%, right 1%

strategy labels: ROT 45%, EARLY 98%, FRONTAL 2%

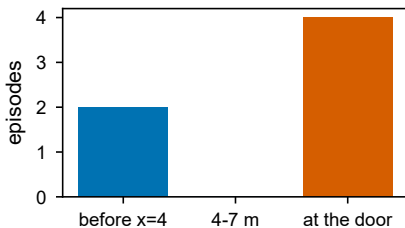
turned at all in 98% of episodes (first turn at x = 2.0 m on average); at apertures needing no turn it turned 98%

pass rate when it inspected itself 89% vs 100% when it never did

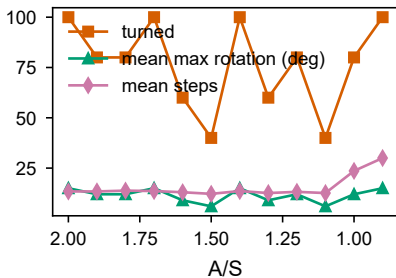
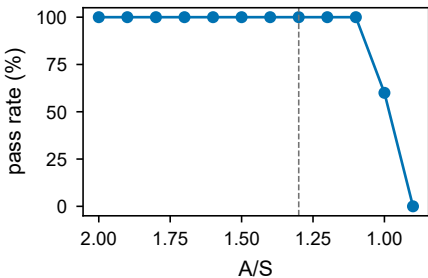
glm-4.6v — 60 episodes, overall pass 92%



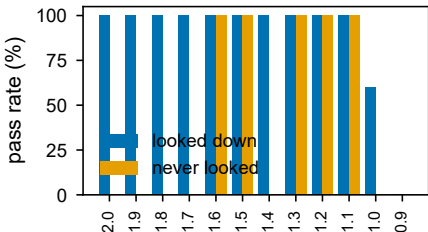
when it looked down (mean $\times$ 5.2 m)

[illegible]

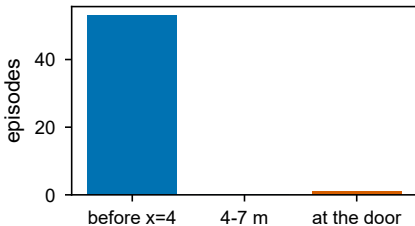
gpt-4.1 — 60 episodes, overall pass 88% behaviour vs aperture



self-inspection (54/60 episodes) and outcome



when it looked down (mean x 1.4 m)



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	100	0	13.4	15	2.4	100	100	—
1.9	1.083	5/5	80	0	13.4	12	2.4	100	100	—
1.8	1.026	5/5	80	0	13.8	12	2.8	100	100	—
1.7	0.969	5/5	100	0	13.6	15	2.6	100	100	—
1.6	0.912	5/5	60	0	13.0	9	2.0	80	100	100
1.5	0.855	5/5	40	0	12.2	6	1.2	60	100	100
1.4	0.798	5/5	100	0	13.6	15	2.6	100	100	—
1.3	0.741	5/5	60	0	12.6	9	1.6	80	100	100
1.2	0.684	5/5	80	0	13.2	12	2.2	80	100	100
1.1	0.627	5/5	80	0	12.6	12	2.2	80	100	100
1.0	0.570	5/5	80	0	12.6	12	2.2	100	60	—
0.9	0.513	5/5	80	0	12.6	12	2.2	100	0	—

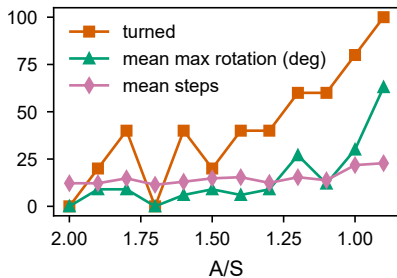
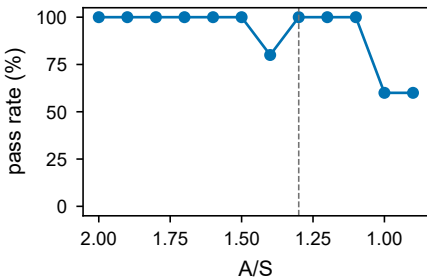
reasoning mentions (share of episodes): opening 100%, align 98%, width 90%, turn 77%, blocked 23%, sideways 23%, look down 13%, turn left 5%, right 2%, left 1%

strategy labels: ROT-EARLY 80%, FRONTAL 23%, SIDEWAYS 7%

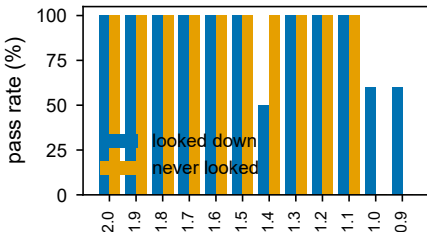
turned at all in 77% of episodes (first turn at x = 1.7 m on average); at apertures needing no turn it turned 100%

pass rate when it inspected itself 87% vs 100% when it never did

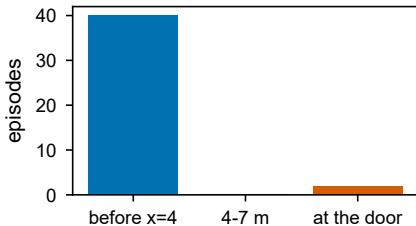
gpt-4o — 60 episodes, overall pass 92% outcome vs aperture behaviour vs aperture



self-inspection (42/60 episodes) and outcome



when it looked down (mean x 1.6 m)



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	12.2	0	1.2	40	100	100
1.9	1.083	5/5	20	20	12.2	9	1.2	60	100	100
1.8	1.026	5/5	40	0	14.8	9	3.8	80	100	100
1.7	0.969	5/5	0	0	11.4	0	0.4	40	100	100
1.6	0.912	5/5	40	0	13.0	6	2.0	60	100	100
1.5	0.855	5/5	20	20	14.8	9	3.8	80	100	100
1.4	0.798	4/5	40	0	15.4	6	4.4	40	50	100
1.3	0.741	5/5	40	0	12.4	9	1.4	80	100	100
1.2	0.684	5/5	60	20	15.4	27	4.4	80	100	100
1.1	0.627	5/5	100	0	13.8	19	2.8	80	100	100
1.0	0.570	3/5	80	0	21.8	30	10.8	100	60	—
0.9	0.513	3/5	100	0	22.8	63	7.8	100	60	—

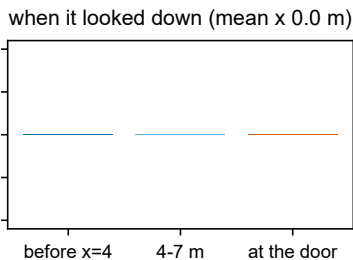
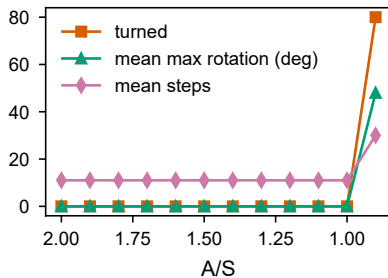
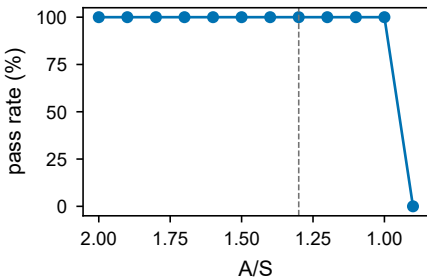
reasoning mentions (share of episodes): opening 88%, width 87%, align 85%, turn 45%, blocked 45%, actions: forward 77%, look down 11%, turn_left 1%, look_left 2%, left 1%

strategy labels: FRONTAL 58%, RCT_EARLY 38%, SIDEWAYS 3%

turned at all in 42% of episodes (first turn at x = 1.6 m on average); at apertures needing no turn it turned 36% of the time

pass rate when it inspected itself 88% vs 100% when it never did

gpt-4o-mini — 60 episodes, overall pass 92% outcome vs aperture behaviour vs aperture



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	5/5	0	0	11.0	0	0.0	0	—	100
1.9	1.083	5/5	0	0	11.0	0	0.0	0	—	100
1.8	1.026	5/5	0	0	11.0	0	0.0	0	—	100
1.7	0.969	5/5	0	0	11.0	0	0.0	0	—	100
1.6	0.912	5/5	0	0	11.0	0	0.0	0	—	100
1.5	0.855	5/5	0	0	11.0	0	0.0	0	—	100
1.4	0.798	5/5	0	0	11.0	0	0.0	0	—	100
1.3	0.741	5/5	0	0	11.0	0	0.0	0	—	100
1.2	0.684	5/5	0	0	11.0	0	0.0	0	—	100
1.1	0.627	5/5	0	0	11.0	0	0.0	0	—	100
1.0	0.570	5/5	0	0	11.0	0	0.0	0	—	100
0.9	0.513	5/5	0	0	11.0	0	0.0	0	—	100

reasoning mentions (share of episodes): align 47%, blocked 8%, turn 7%

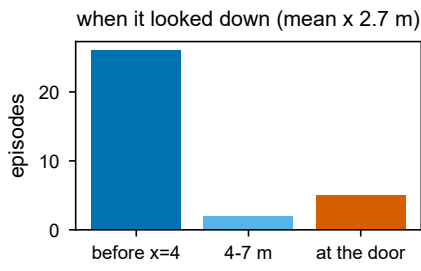
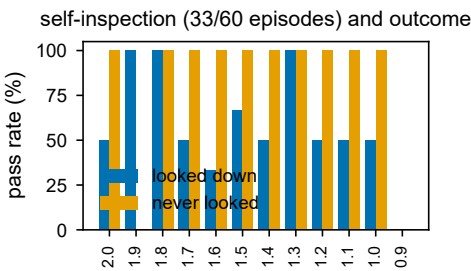
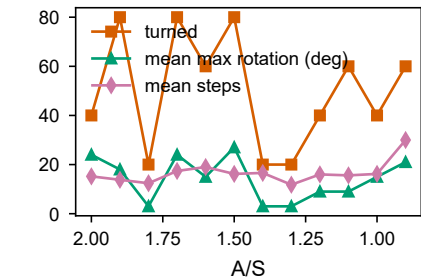
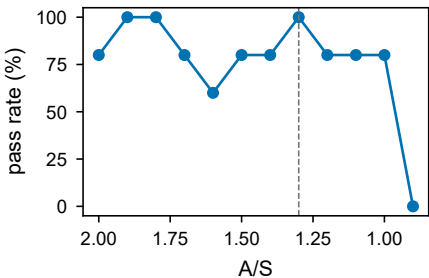
actions: forward 90%, left 4%, right 3%, turn_left 1%, turn_right 1%

strategy labels: FRONTAL 92%, ROT_LATE 7%, SIDEWAYS 2%

turned at all in 7% of episodes (first turn at x = 7.2 m on average); at apertures needing no turn it turned 0%

pass rate when it inspected itself 0% vs 92% when it never did

gwen-vl-max — 60 episodes, overall pass 77% outcome vs aperture behaviour vs aperture



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	40	0	15.2	24	4.2	40	50	100
1.9	1.083	5/5	80	0	13.8	18	2.8	100	100	—
1.8	1.026	5/5	20	0	12.4	3	1.4	80	100	100
1.7	0.969	4/5	80	0	17.4	24	6.4	40	50	100
1.6	0.912	3/5	60	0	19.0	15	8.0	60	33	100
1.5	0.855	4/5	80	0	16.2	27	5.2	60	67	100
1.4	0.798	4/5	20	0	16.6	3	5.6	40	50	100
1.3	0.741	5/5	20	0	11.8	3	0.8	20	100	100
1.2	0.684	4/5	40	0	16.0	9	5.0	40	50	100
1.1	0.630	4/5	40	0	16.2	21	5.0	40	50	100
1.0	0.570	4/5	40	0	16.2	19	5.2	40	50	100

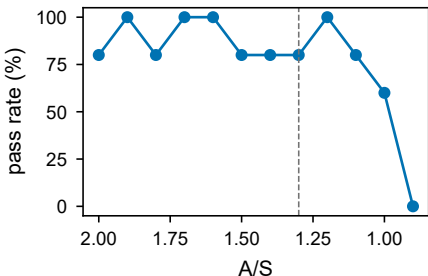
reasoning mentions (share of episodes): align 100%, turn 67%, opening 65%, width 63%, blocked 60%, sideways 24%, look down 9%, turn_left 4%, right 3%, turn_right 3%

strategy labels: FRONTAL 47%, ROT_EARLY 40%, SIDEWAYS 8%, ROT_MID 3%, ROT_LATE 2%

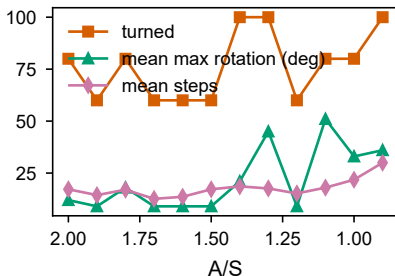
turned at all in 50% of episodes (first turn at x = 2.5 m on average); at apertures needing no turn it turned at 49%

pass rate when it inspected itself 58% vs 100% when it never did

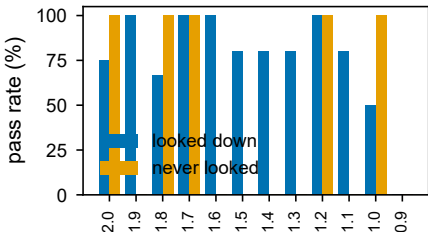
qwen3-vl-235b-a22b-instruct — 60 episodes, overall pass 78%
outcome vs aperture



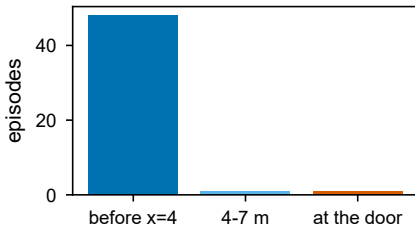
behaviour vs aperture



self-inspection (50/60 episodes) and outcome



when it looked down (mean x 1.4 m)



made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	80	0	17.2	12	6.2	80	75	100
1.9	1.083	5/5	60	0	14.4	9	3.4	100	100	—
1.8	1.026	4/5	80	0	17.0	18	6.0	60	67	100
1.7	0.969	5/5	60	0	12.6	9	1.6	20	100	100
1.6	0.912	5/5	60	0	13.6	9	2.6	100	100	—
1.5	0.855	4/5	60	0	17.2	9	6.2	100	80	—
1.4	0.798	4/5	100	0	18.6	21	7.6	100	80	—
1.3	0.741	4/5	100	0	17.6	45	6.6	100	80	—
1.2	0.684	5/5	60	0	15.2	9	4.2	80	100	100
1.1	0.627	5/5	100	0	14.9	33	1.9	100	80	—
1.0	0.570	3/5	60	0	11.8	33	2.6	80	50	100

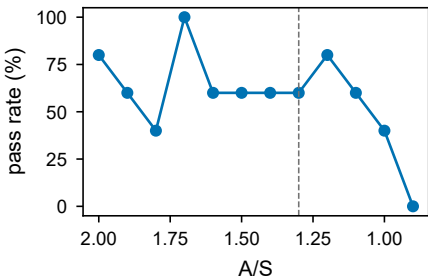
reasoning mentions (share of episodes): align 100%, opening 97%, turn 97%, width 93%, blocked 95%, sideways 38%, look down 7%, turn_right 7%, turn_left 6%, look left 4%

strategy labels: ROT_EARLY 73%, FRONTAL 23%, ROT_LATE 2%, SIDEWAYS 2%

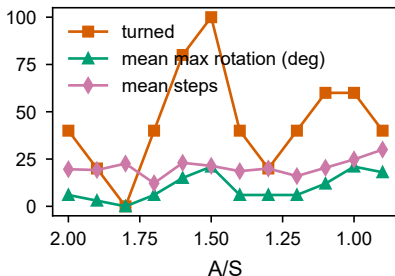
turned at all in 77% of episodes (first turn at x = 1.8 m on average); at apertures needing no turn it turned 100%

pass rate when it inspected itself 76% vs 90% when it never did

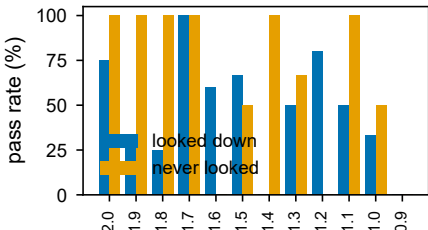
qwen3-vl-32b-instruct
outcome vs aperture



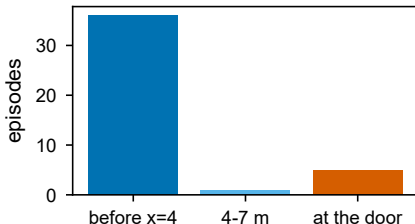
60 episodes, overall pass 58%
behaviour vs aperture



self-inspection (42/60 episodes) and outcome



when it looked down (mean x 1.7 m)



s made sideways; pass|look / pass|nolook = pass rate within that aperture when it did or did not inspect its body;

A/S	width	pass	turn%	side%	steps	maxrot	excess	look%	pass look	pass nolook
2.0	1.140	4/5	40	0	19.6	6	8.6	80	75	100
1.9	1.083	3/5	20	0	19.2	3	8.2	60	33	100
1.8	1.026	2/5	0	0	22.6	0	11.6	80	25	100
1.7	0.969	5/5	40	0	12.2	6	1.2	40	100	100
1.6	0.912	3/5	80	0	23.0	15	12.0	100	60	—
1.5	0.855	3/5	100	0	21.4	21	10.4	60	67	50
1.4	0.798	3/5	40	0	18.6	6	7.6	40	0	100
1.3	0.741	3/5	20	0	20.0	6	9.0	40	50	67
1.2	0.684	4/5	40	0	16.0	6	5.0	100	80	—
1.1	0.627	2/5	80	0	14.8	21	13.8	80	50	100
1.0	0.570	2/5	40	0	12.8	21	13.8	100	0	—

reasoning mentions (share of episodes): align 100%, opening 87%, width 83%, blocked 89%, turn 88%, sideways 32%, look down 9%, right 5%, left 4%, look_left 3%

strategy labels: FRONTAL 43%, ROT_EARLY 32%, SIDEWAYS 23%, ROT_LATE 2%

turned at all in 45% of episodes (first turn at x = 1.2 m on average); at apertures needing no turn it turned 49%

pass rate when it inspected itself 48% vs 83% when it never did